Fiscal Year:	FY 2024	**Task Last Updated:**	FY 12/26/2024
PI Name:	Temple, David Ph.D.		
Project Title:	Systematic Assessment of Noisy Galvanic Vestibular Stimulation as a Sensorimotor Countermeasure		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80303-7813	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2022 TRISH-RFA-2201-PD: Translational Research Institute for Space Health (TRISH) Postdoctoral Fellowships
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No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	6	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Clark, Torin Ph.D. (MENTOR: University of Colorado, Boulder)		
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Task Description:	POSTDOCTORAL FELLOWSHIP Here, we propose noisy Galvanic Vestibular Stimulation (nGVS) as a potential countermeasure to mitigate sensorimotor disruptions experienced by astronauts during gravity transitions. nGVS is thought to improve sensory information throughput via a phenomenon known as stochastic resonance (SR). Unfortunately, several gaps in the literature exist that prevent the utilization of nGVS as a viable countermeasure for astronauts. We aim to systematically assess the efficacy of utilizing nGVS as a countermeasure to improve astronaut performance through: 1) optimizing the waveform of the noisy electrical current for vestibular enhancement, 2) determining if optimal vestibular enhancement generalizes to optimal performance on a number of balance, locomotor, and manual control tasks, 3) investigating potential after-effects of the nGVS on performance, and 4) testing the effects of nGVS in a gravity transition analog, with the goal of increasing the countermeasure readiness level. We propose to accomplish these objectives through a series of human subject experiments systematically testing various nGVS waveforms' impact on vestibular perceptual thresholds to determine the optimal waveform for inducing SR within the vestibular system (Aim 1). Then we will implement the optimal vestibular perceptual nGVS waveform in various balance, locomotor, and manual control nulling tasks to determine if relationships exist between SR-induced optimal vestibular performance and optimal performance in various operational tasks performed by astronauts (Aim 2). Next, we will assess vestibular perceptual thresholds and balance performance metrics prior to, during, immediately after, and 1 hour after prolonged exposure to nGVS to evaluate any potential aftereffects nGVS might have on the functional performance (Aim 3). Finally, we will use a ground-based analog for gravity transitions relevant for astronauts, which applies sustained hyper-gravity centrifugation (Aim 4). This analog induces sensorimotor disturbances following centrifugation, similar to post-flight astronauts, and we will test whether the optimized nGVS application can mitigate balance impairment. If successful, this research effort will deliver a validated countermeasure for astronaut sensorimotor impairment due to gravity transitions. Critically, nGVS is a low-cost, mass, power, and volume approach that is safe, well-tolerated, and could be utilized during long-duration exploration in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	A machine learning algorithm was developed that assesses the likelihood of SR exhibition from multiple vestibular thresholds assessed at various nGVS amplitudes. This marks an improvement and more objective means of identifying SR over previous methods, such as utilizing judges. Furthermore versions of this algorithm were developed when computing vestibular thresholds with a Confidence Signal Detection model and fewer trials (30, 40, 50, 60, 70, 80, and 90) than are traditionally needed with binary thresholds (at least 100 trials). Computed vestibular thresholds with a CSD model were shown to exhibit SR in a similar manner to traditional binary thresholds, allowing future research to be more efficient and accurate when assessing SR induced vestibular threshold improvements. We have submitted a manuscript to Experimental Brain Research describing our methods and findings for the above work in hopes of disseminating this information to a larger group of individuals who might have interest in inducing vestibular enhancement with nGVS. Such research has the potential to assist a wide range of individuals with vestibular impairment (e.g., elderly, clinical populations, traumatic brain injury, etc.) in addition to potentially assisting astronauts with post-flight sensorimotor impairment. GVS stimulators are very low mass, power (e.g., typically less than 1 mA of current draw), volume, user-friendly, and can be easily utilized in microgravity or terrestrial (i.e., Earth, Lunar, and/or Martian) environments, making them a feasible means of modulating vestibular afferents in a variety of scenarios. Our findings of SR-induced enhancement of vestibular thresholds with nGVS continue to support previous literature suggesting its use as a potential countermeasure. However, our findings do not appear to be consistent with those observing benefits after the nGVS is turned off (after-effects). It is possible that some prior studies not utilizing a sham condition could have mistaken task learning effects as beneficial nGVS after-effects, which is why our study was designed to include a sham condition. To effectively improve performance in the tasks we assessed, it appears that the nGVS may need to be concurrently running.
Bibliography Type:	Description: (Last Updated: 03/12/2025)
Abstracts for Journals and Proceedings	Lonner T, Allred A, Gopinath A, Mogheim T, Temple D, Clark T. "Safe return: Mitigating motion sickness for Earth re-entry and crew egress." Aerospace Medical Association 94th Annual Scientific Meeting, Chicago, IL, May 6-9, 2024. AsMA 2024 Meeting Abstracts. Aerospace Medical Association 94th Annual Scientific Meeting, Chicago, IL, May 6-9, 2024. , May-2024
Abstracts for Journals and Proceedings	Temple DR. "Potential spaceflight applications for galvanic vestibular stimulation." International Space Development Conference, Frisco, TX, May 25-28, 2023. Abstracts. International Space Development Conference, Frisco, TX, May 25-28, 2023. , May-2023
Abstracts for Journals and Proceedings	Temple DR, Stone T, Clark TK. "Enhancement of vestibular perceptual thresholds with noisy galvanic vestibular stimulation: Implications as a sensorimotor countermeasure." 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. , Feb-2024
Articles in Peer-reviewed Journals	Stone T, Clark TK, Temple DR. "Noisy galvanic vestibular stimulation induces stochastic resonance in vestibular perceptual thresholds assessed efficiently using confidence reports." Exp Brain Res. 2024 Dec 24;243(1):34. https://doi.org/10.1007/s00221-024-06984-8 ; PMID: 39718639 , Dec-2024